Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC Accredited with 'A++' Grade.
Ph:07104-242919, 242623,242588 Website: www.ycce.edu, Email: principal@ycce.edu

Department of Applied Chemistry (2024-25)

NOTES

UNIT-V

Spectroscopic Techniques and Applications

Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
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Ph:07104-242919, 242623,242588 Website: www.ycce.edu, Email: principal@ycce.edu

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Unit V: Spectroscopic Techniques and Applications

Syllabus:
Introduction, fundamentals, types, principles, and selection rules of spectroscopy.
Basic principle and applications of UV- Visible, IR, NMR Spectroscopy and numericals.

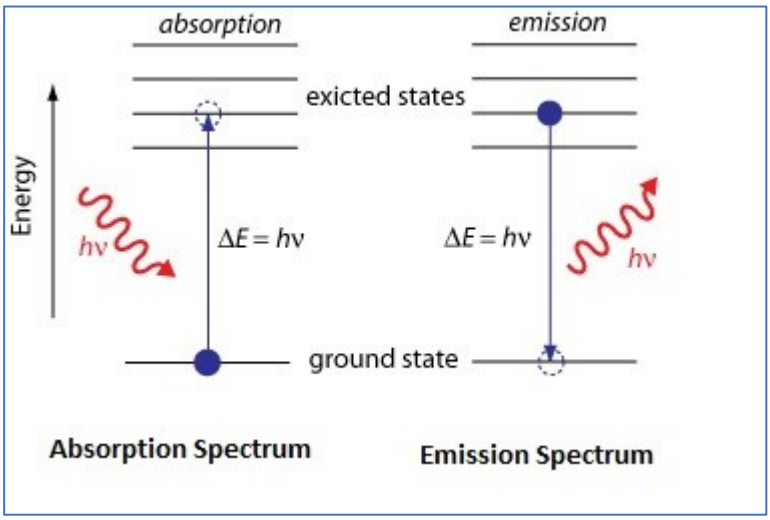
Introduction

Spectroscopy is the branch of science dealing with the study of interaction of electromagnetic radiation with matter like atoms and molecules. Spectroscopy gives a lot of information about the object under study. It deals with the transition that an electron in a molecule undergoes between energy levels upon absorption of suitable radiations that is used to determine atomic and molecular structure.
Atoms and molecules interact with electromagnetic radiation (EMR) and absorb and/or emit Electromagnetic Radiation. Absorption of EMR stimulates different types of motion in atoms and/or molecules. The patterns of absorption (wavelengths absorbed and to what extent) and/or emission (wavelengths emitted and their respective intensities) are called 'spectra'

Consider a molecule having two energy levels, say E_1 and E_2.

Absorption Spectrum results when an electron in an atom or molecule undergoes a transition from the energy level E_1 (ground state) to energy level E_2 (excited state) by absorption of photon having energy ($E=h\nu$), provided this energy is exactly equal to energy difference ($\Delta E = E_2 - E_1$) between the two levels as shown in figure.

Emission spectrum results when an electron in an atom or molecule falls from the excited state to the ground state with the emission of a photon of energy ($h\nu$) as shown in figure.



Aim of spectroscopy: Determination of structure of unknown sample (for which structure not yet established)

Advantages of spectroscopy

1) Little quantity of sample enough for analysis
2) Simple and Short time method
3) Sample can be recovered (except in mass) without chemical reaction on sample
4) More accurate method, absolute structure can be obtained but expensive technique.

Important Term Involved In Specroscopy

1) Electromagnetic radiations (EMR):-

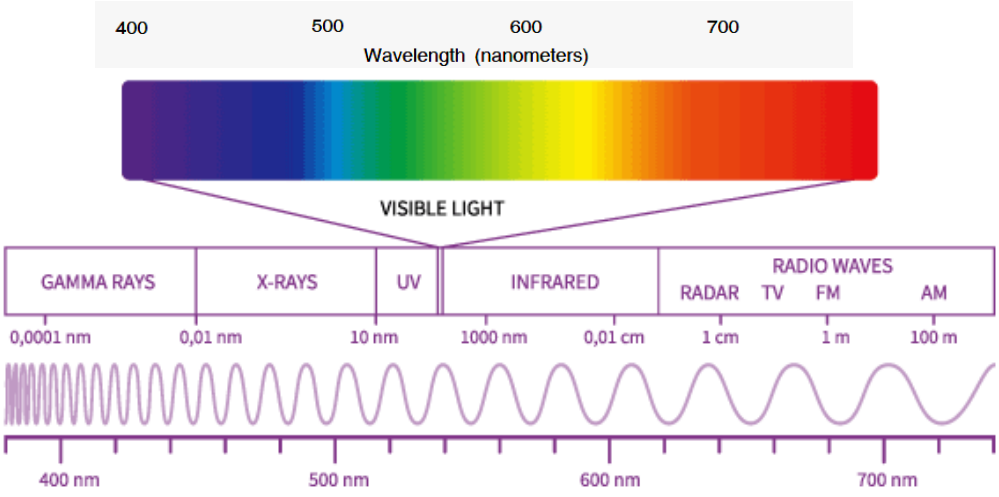
- EMR is one kind of energy
- EMR have dual nature i.e. wave and particle nature
- EMR is quantized source of energy which is required for spectroscopy
- EMR radiation associated with oscillatory electric and magnetic field.

Example of EMR is IR radiation, UV-visible radiation, radiofrequency radiation, x-ray radiation etc.

NOTE: Spectroscopy is considered only as a quantised energy.

2) Electromagnetic Spectrum

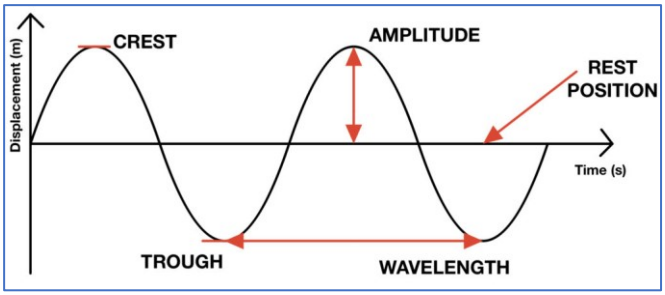
- The arrangement of all types of electromagnetic radiations in order of their increasing wavelengths or decreasing frequencies is known as complete electromagnetic spectrum.
- Wavelength above the visible region is called Infra-red while that below it is the ultra-violet region.
- Infra-red radiations have longer wavelengths and are thus, less energetic. Cosmic rays carry high energy while radiowaves are least energetic. Microwaves have larger wavelengths and are used in telephone transmission.



2) **Wavelength (λ):**

Distance between adjacent crest or trough in a wave is called as wavelength.
It is denoted by λ and expressed in micrometer, nanometer, angstorm
$1\mu m=10^{-6}m$
$1nm= 10^{-9}m$

$1A^0 = 10^{-10}$ m



3) Amplitude: (A)
Maximum height of crest or depth of trough is called as amplitude. It is used to measure intensity or brightness. More the amplitude, more will be the intensity or brightness and vice versa.

4) Velocity: (c)
"Linear Distance travelled by wave in one second is called as velocity."
All electromagnetic waves travel with the same speed of $3x10^8 ms^{-1}$ in vacuum, whatever may be the frequency or wavelength.

Velocity (c)= Distance travelled by wave in one second
= Wavelength **(λ)** x Frequency **(ν)**
$c = \lambda \nu$

5) Frequency: (ν)
"Number of wave passing through a given point in one second"
In Short Number of waves per unit time.
UNIT: Hz, cycle /sec
1Hz = 1 cycle/sec $\quad \nu = c/\lambda$

6)Wavenumber:$\bar{\nu}$
"Number of wave present in unit length"
Wavenumber is reciprocal of wavelength, expressed as cm^{-1}, m^{-1}

$$\bar{\nu} = 1/\lambda$$
$$\nu = c/\lambda$$
$$\nu = c.\bar{\nu}$$

i.e. frequency directly proportional to wavenumber

7) Energy: Energy possessed by a wave in cal/mol
$E = h\nu$
$= h.c/\lambda$ h= Plancks Constant=$6.625x10^{-34}$Js

i.e. Energy is inversely proportional to the wavelength, wavelength increases and energy decreases.
In UV energy is sufficient for the excitation of electron.
In IR the energy is slightly more than radiofrequency and less than visible which can cause the vibration of the atom.
In NMR radiofrequency is used which has very minimum energy which can cause the resonance of nucleus only.

Absorption of Electromagnetic Radiation:

If some light radiation is passed through a sample of an organic compound, then some of the wavelengths are absorbed while others belonging to that light source remain unaffected. A molecule can only absorb radiation of certain frequency, if there exists within a molecule an energy transition of magnitude, ΔE which is equal to $h\nu$. It is found that when light radiations are passed through an organic compound, then electrons of the component atoms are excited. In addition, the vibrational and the rotational energies of the molecules as a whole are quantised. Thus, any wavelength of light that a particular molecule will absorb will be due to the changes in the electronic, vibrational or rotational energy levels permissible for its atoms.

The interaction of EMR with matter gives rise to two types of spectra namely atomic spectra and molecular spectra.

1. **Atomic spectrum** arises from the transition of electrons from one energy level to another due to changes of energy in the atom. It gives detailed information about electronic structure of atoms.
2. **Molecular spectrum** involves transition between rotational and vibrational energy levels in addition to electronic transition.
   Molecular spectrum is much more complicated than the atomic spectrum. It provides most useful information regarding the shape and size of molecules, the bond angles, bond lengths, strength of bonds and bond dissociation energies. Hence molecular spectroscopy is of great use in determining the structure and constitution of compounds.

The wavelengths absorbed are measured with the help of a **spectrometer**. If we plot the changes in absorption against wavelengths, we get certain absorption bands which are highly characteristic of a compound and the technique provides an excellent tool to ascertain the molecular structure of an unknown substance.

Different forms of spectroscopy

Spectroscopy is a fundamental tool of scientific study, with applications ranging from materials characterization to astronomy and medicine. Spectroscopy techniques are commonly categorized according to the wavelength region used, the nature of the interaction involved, or the type of material studied.

Except X-ray spectroscopy no single spectroscopy provides complete (full) structure of the compound. All though X-ray spectroscopy gives full structure but con not be used for all samples because x-ray spectroscopy required perfectly crystalline sample. Hence spectroscopy divided into different type depending upon the energy source and application.

Type of Spectroscopy	Usual Wavelength Range	Transition	Sensitive compounds
X-ray diffraction	0.1-100 Å	Inner electron	Crystalline compounds
Ultraviolet visible absorption	180 -780 nm (UV-Visible)	Bonding electrons Transition	Double conjugated bonds, double, triple, peptides, aromatic carbonyl compounds
Infrared absorption	0.78-300 μ m (infrared)	Rotation/vibration of molecules (changes in dipole moment)	Polar bonds (N-H, C-H, O-H, S-H, C-O)

Raman scattering	0.78-300 μ m (Far infrared)	Rotation/vibration of molecules (changes in polarazibility)	Nonpolar double or triple bonds (C≡C, C=C)
Microwave absorption	0.75-3.75 mm (Microwaves)	Rotation of molecules	Molecules must have permanent dipole (HCl, H2O)
Electron spin resonance	3 cm (Microwaves)	Electron spin changes in a magnetic field	Presence of unpaired electron in molecule
Nuclear magnetic resonance	0.6-10 m (Radio waves)	Nuclear Spin changes in a magnetic field	Nuclei having spin quantum number I>1

Selection Rule

- The probability of transition between two energy levels is determined by Selection rules.
- Atomic and molecular species contain a very large number of states in which energy can be distributed, although generally only the states lying lowest in energy will be thermally populated.
- If electromagnetic radiation could effectively stimulate transitions between any one of these states, atomic and molecular spectra would be complex.
- Selection rules, limit the possible transitions and render these spectra amenable to analysis.
- On the basis of selection rules transitions are divided in two categories:
  Allowed transitions: which obey selection rules
  Forbidden transitions: which violate selection rules
- The selection rules may differ according to the technique used to observe the transition.
- Selection rules are obtained from quantum theory of interaction of radiation with matter, which forms the backbone of spectroscopy.

Sr no.	Transition	Selection Rule	Remarks
1	Electronic	a. $\Delta S = 0$ The Spin Rule	S–Spin Quantum Number Transitions that are permitted must involve an electron promotion without a change in its spin
		b. $\Delta l = +/-1$ The Orbital Rule or Laporte	l- Angular Quantum Number Transitions inside a specific set of p or d orbitals are prohibited if the molecule contains a centre of symmetry
2	Vibrational	$\Delta v = \pm 1$	*v- Vibrational Quantum number* Transitions that are permitted must have vibrational quantum number change by unity
3	Rotational	$\Delta J = \pm 1$	*J- Rotational Quantum number* Transitions that are permitted must have rotational quantum number change by unity

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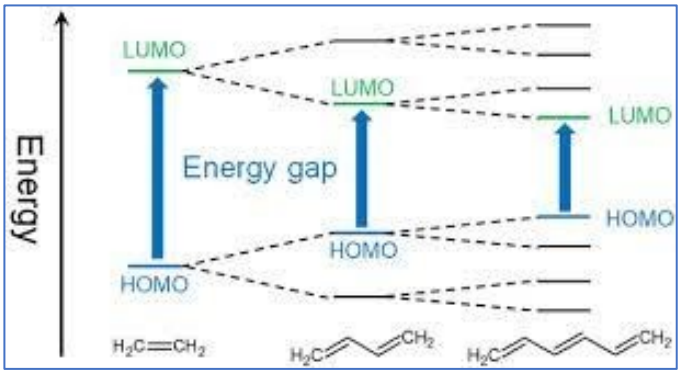
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ULTRAVIOLET (UV) & VISIBLE SPECTROSCOPY

Ultraviolet and visible spectroscopy also known as electronic spectroscopy is used to measure the number of conjugation of double bond and aromatic conjugation within the molecule.

It involves the promotion of electrons from HOMO to LUMO (HOMO means Highest Occupied Molecular Orbital whereas LUMO means Lowest Unoccupied Molecular Orbital).

The HOMO-LUMO gap decrease as the conjugation increase.



The ultraviolet region corresponds to 400-200 nm and visible region to 800- 400 nm.

UV-visible spectroscopy based on **Beer-Lambert law**.

$$\log {}^{I_o}/_{I} = \varepsilon . l . c \text{ or } \epsilon = A/c\,l$$

Where,

I_o = intensity of incident light

I = intensity of transmitted light

ε = molar absorptivity

***l* = path length of sample**

c = concentration of sample

A = Absorbance

Absorbance (A): It is reciprocal of Transmittance

$$\text{Absobance (A)} = \text{Optical density (D)} = \log \frac{1}{T} \text{ or } \log \frac{I_o}{I}$$

Transmittance (T): The fraction of incident light transmitted is known as Transmittance.

$$\text{Transmittance (T)} = \frac{I}{I_o}$$

Principle of UV-Visible Spectroscopy

The Principle of UV-Visible Spectroscopy is based on the absorption of ultraviolet light or visible light by chemical compounds, which results in the production of distinct spectra

When matter absorbs ultraviolet radiation, the electrons present in it undergo excitation. This causes them to jump from a ground state (an energy state with a relatively small amount of energy associated with it) to an excited state (an energy state with a relatively large amount of energy associated with it). It is important to note that the difference in the energies of the ground state and

the excited state of the electron is always equal to the amount of ultraviolet radiation or visible radiation absorbed by it.

The Absorption Laws

The Beer-Lambert law is known by so many names because more than one law is involved.

- In 1729, Pierre Bouguer discovered the law.
- Later, in 1760, Johann Heinrich Lambert quoted Bouger's discovery saying that the absorbance of a sample is directly proportional to the path length of light. Although Lambert dint claim the discovery, he was often credited with it.
- In 1852, August Beer discovered a related law which stated that the absorbance is proportional to the concentration of the sample.

So, there are two laws which govern the absorption of light by the molecules. These are:
Lambert's law and Beer's law

1) **Lambert's Law:** It states that: *When a beam of monochromatic radiation passes through a homogeneous absorbing medium, the rate of decrease of intensity of radiation with thickness of absorbing medium is proportional to the intensity of the incident radiation.*
2) **Beer's Law:** This law states that: *When a beam of monochromatic radiation is passed through a solution of an absorbing substance, the rate of decrease of intensity of radiation with thickness of the absorbing solution is proportional to the intensity of incident radiation as well as the concentration of the solution.*

On combining the two laws, the Beer-Lambert Law can be formulated as below:

$$\log \frac{I_0}{I} = \epsilon . c . l = A$$

Where

I_0 = Intensity of incident light
I = Intensity of transmitted light
c = Concentration of solution in moles litre–1
l = Path length of the sample (usually 1 cm)
ϵ = Molar extinction coefficient (or molar absorptivity)
A = Absorbance

Beer-Lambert's Law Applications

This law finds applications in various fields such as:

1) *Analytical chemistry:* This analysis mainly concentrates on the separation, quantification, and identification of matter by spectrophotometry. There is no involvement of extensive pre-processing of the sample to get the results. For example, bilirubin count in a blood sample can be determined by using a spectrophotometer.
2) In atmosphere Solar or stellar radiation in the atmosphere ***can be described using this law.***

Limitations of Beer Lambert's Law : This law is not obeyed

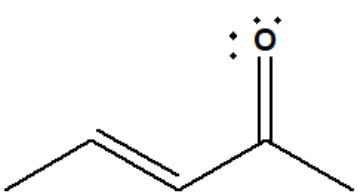
1) When different forms of the absorbing molecules are in equilibrium as in keto-enol tautomers.
2) When fluorescent compounds are present.
3) When solute and solvent form complexes through some sort of association.

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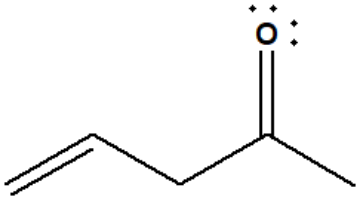
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Application of UV spectroscopy

1) **For characterization of aromatic compounds and conjugated olefins.**
2) **Extent of conjugation**: As the double bond increase, absorption shift towards longer wavelength. The extent of conjugation in polyenes R—(CH=CH)*n*—R can be estimated. With the increase in the number of double shifts the absorption to longer wavelength.
3) Distinction between conjugated and non-conjugated compounds:



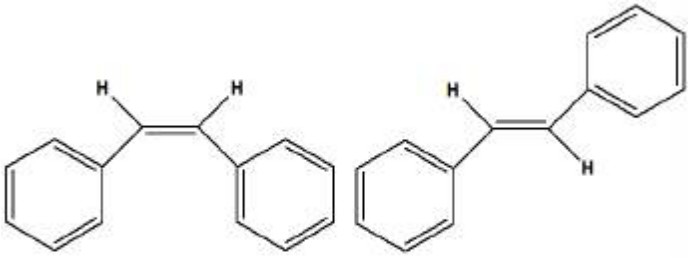
A



B

Both the above compound A & B are isomer of each other. Due to $n - \pi^*$transition, compound A appear at longer wavelength than compound B.

4) **Detection of functional groups**: The technique is applied to detect the presence or absence of the chromophore (group imparting color ex. Nitro group). The absence of a band at a particular wavelength may be regarded as an evidence for the absence of a particular group in the compound.
5) **Determination of Concentration:** The variation of color with concentration forms the basis of colorimetry. Unknown concentration of coloured solutions can be determined by using UV-visible radiation of proper wavelength and Beers Lamberts Law.
6) **Used in the detection of impurities**: Due to presence of impurities in the compound, additional peaks are observed in UV spectrum.
7) **Identification of an unknown compound**: An unknown compound can be identified by comparing its spectrum with the known spectra. If the two spectra coincide, the two compounds must be identical. If the two spectra do not coincide, then the expected structure is different from the known compound.
8) **Study of kinetis of chemical reaction**: it is done by studying the change in concentration of reactants or products with time, provided reactants or products exhibit absorbtion in UV-Visible range.
9) **Study of Geometrical isomers**. The results of absorption show that cis-alkenes absorb at different wavelengths as compared to their corresponding trans isomers.



A (cis form)
λ_{max} = 283 nm

B (trans form)
λ_{max} = 296 nm

In isomer A, due to steric hindrance, absorption shift towards lower wavelength. We know that, steric hindrance destabilize the compound means energy of the molecule increase which decreases the wavelength.

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INFRARED (IR) SPECTROSCOPY

VIBRATIONAL SPECTROSCOPY

IR spectroscopy (which is short for infrared spectroscopy) deals with the infrared region of the electromagnetic spectrum, i.e. light having a longer wavelength and a lower frequency than visible light. Infrared Spectroscopy is based on the fact that interatomic bonds of a molecule interact with infrared light.

Regions of the Infrared spectrum

Functional Group Region- 4000 cm^{-1} to 1300 cm^{-1}: Most of the bands that indicate what functional group is present are found in this region. Their bands can be identified and used to determine the functional group of an unknown compound.

Finger Print Region- 1300 cm^{-1} to 400 cm^{-1}**:** Bands that are unique to each molecule, similar to a fingerprint, are found in the fingerprint region, from. These bands are only used to compare the spectra of one compound to another.

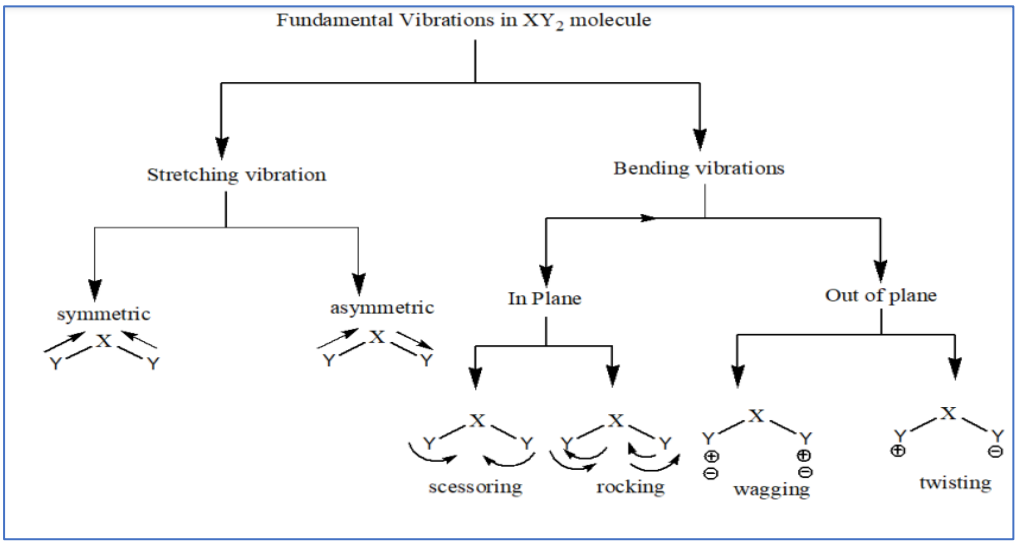


Principle of Infrared Spectroscopy

The IR spectroscopy theory utilizes the concept that molecules tend to absorb specific frequencies of light that are characteristic of the corresponding structure of the molecules. The energies are reliant on the shape of the molecule, the associated vibronic coupling, and the mass corresponding to the atoms.

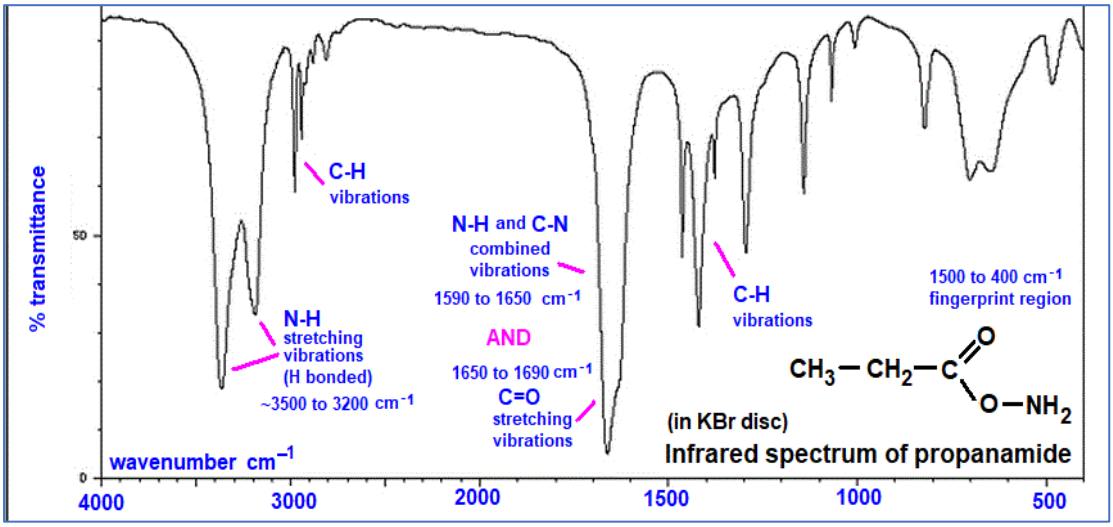
The energy required to excite the bonds belonging to a molecule, and to make them vibrate with more amplitude, occurs in the Infrared region.

For instance, the molecule can absorb the energy contained in the incident light and the result is a faster rotation or a more pronounced vibration. Covalent bonds can vibrate in several modes, including stretching, rocking, and scissoring.



When molecule absorbed electromagnetic radiation in IR region, it undergoes vibrational or a rotational transitions which causes net change in the dipole moment in the molecule. We know that each vibrational level is associated with a number of closely spaced rotational levels. Clearly, the Infra-red spectra is considered as **vibrational-rotational spectra.**

All the bonds in a molecule are not capable of absorbing infra-red energy but only those bonds which are accompanied by **a change** in **dipole moment** will absorb in the infra-red region. A bond will only interact with the electromagnetic infrared radiation, if it is polar. Such vibrational transitions which are accompanied by a change in the dipole-moment of the molecule are called infra-red active transitions. Thus, these are responsible for absorption of energy in the Infra-red region. On the other hand, the vibrational transitions which are not accompanied by a change in dipole-moment of the molecule are IR inactive (for example: O_2, H_2, N_2 etc.) and does not absorb IR radiation. For example, vibrational transitions of C = O, N—H, O—H etc. bands are accompanied by a change in dipole-moment and thus, absorb strongly in the Infra-red region. But transitions in Carbon-Carbon bonds in symmetrical alkenes and alkynes are not accompanied by the change in dipole-moment and hence do not absorb in the infra-red region. Symmetrical non-polar bonds in N≡N and O=O do not absorb radiation, as they cannot interact with an electric field. It is important to note that since the absorption in infra-red region is quantised, a molecule of the organic compound will show a number of peaks in the infra-red region. If dipole moment does not change in molecules then they are IR inactive (for example: O2, H2, N2 etc.) means they does not absorb IR radiation.



The major use of infrared spectroscopy is to determine the functional groups of molecules, relevant to both organic and inorganic chemistry. An IR spectrum is essentially a graph plotted with the infrared light absorbed on the Y-axis against frequency or wavelength on the X-axis. IR Spectroscopy detects frequencies of infrared light that are absorbed by a molecule. Molecules tend to absorb these specific frequencies of light since they correspond to the frequency of the vibration of bonds in the molecule.

Application of IR Spectroscopy

1) **Identification of different functional group**: The presence or the absence of absorption bands help in predicting the presence of certain functional groups in the compound.
2) **Identification of unknown compound**: It is done by finger print technique, which involves matching the IR spectra of unknown compound with standard database of known compounds.

3) **Distinction between two types of hydrogen bonding:** intermolecular and intra-molecular hydrogen bonding can be distinguished as both give absorption at different frequency.
4) **Identification of purity of the compound**: if compound is impure then additional peaks are observed in the IR spectrum.
5) **Quantitative analysis**: It helps to determine amount of components in a organic mixture as the size of peak is a direct indication of the amount of material present.

Table: IR correlation chart

	Types of vibration	Frequency (cm^{-1})
Alkane	C-C stretching	1200
	C-H stretching	3000-2840
	-CH2 bending	1465
	-CH3 bending	1375
	CH2 rocking	720
Alkenes	C=C stretching	1650
	=C-H stretching	3095-3010
	=C-H bending	1000-650
Alkynes	C≡C stretching	2100
	≡C-H stretching	3300
	≡C-H bending	700-600
Aromatic	C=C stretching	1600, 1500, 1450
	=C-H stretching	3040-3010
	=C-H bending	Below 900
C=O stretching	Amide	1680
	Carboxylic acid	1710
	Ketone	1715
	Aldehyde	1725
	Ester	1735
	Acid chloride	1800
	Anhydride	1760 (I), 1810 (II)
C-O	Stretching	1300-1000
O-H	Alcohol, phenol	
	Free	3600
	H-bonded	3400-3200
	Carboxylic acid	3400-2400
Amine	-N-H stretching	3440 (as), 3350 (s)
	-N-H bending	1650-1580
	C-N stretching	1350-1000
Amide	-N-H stretching	3370 (as), 3150 (s)
	-N-H bending	1650-1560
Nitriles	C≡N	2250
Imines	C=N	1690-1640

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Nuclear Magnetic Resonance Spectroscopy

Nuclear magnetic resonance (NMR) spectroscopy is the study of molecules by recording the interaction of radiofrequency (Rf) electromagnetic radiations with the nuclei of molecules placed in a strong magnetic field.
Zeeman first observed the strange behaviour of certain nuclei when subjected to a strong magnetic field at the end of the nineteenth century, but the practical use of the so-called ***"Zeeman effect"*** was only made in the 1950s when NMR spectrometers became commercially available.

Nuclear Magnetic Resonance (NMR) was first detected experimentally at the end of 1945, nearly concurrently with the work groups Felix Bloch, Stanford University and Edward Purcell, Harvard University. The first NMR spectrum was first published in the same issue of the Physical Review in January 1946. Bloch and Purcell were jointly awarded the 1952 Nobel Prize in Physics for their research of Nuclear Magnetic Resonance Spectroscopy.

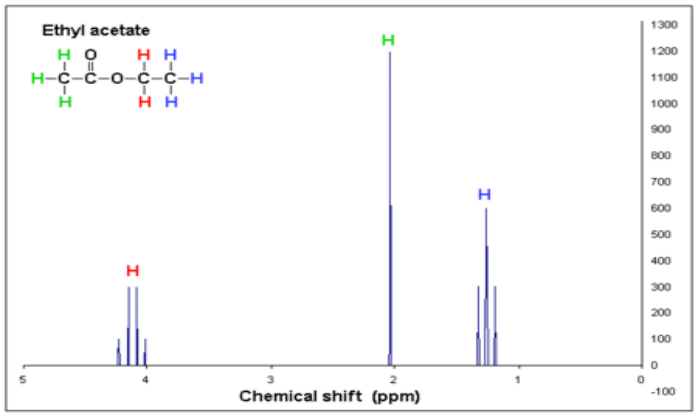
Nuclear magnetic resonance (NMR) spectroscopy is a crucial analytical tool for organic chemists. The research in the organic lab has been significantly improved with the aid of the NMR. Not only can it provide information on the structure of the molecule, it can also determine the content and purity of the sample. Proton (1H) NMR is one of the most widely used NMR methods by organic chemists. The protons present in the molecule will behave differently depending on the surrounding chemical environment, making it possible to elucidate their structure.

NMR Spectroscopy Principle

- NMR is a branch of spectroscopy in which radio frequency waves induce transitions between magnetic energy levels of nuclei of a molecule. The magnetic energy levels are created by keeping the nuclei in a magnetic field.
- Many nuclei have spin, and all nuclei are electrically charged, according to the NMR principle.
- In some atoms (eg ^{12}C, ^{16}O, ^{32}S) these spins are paired and cancel each other out so that the nucleus of the atom has no overall spin. However, in many atoms (^{1}H, ^{13}C, ^{31}P, ^{15}N, ^{19}F etc) the nucleus does possess an overall spin.
- The nuclei with spin quantum number (I) greater than zero can exhibit NMR phenomenon.
- Transfer of energy is possible from base energy to higher energy levels when an external magnetic field is applied when nuclei are electrically charged and have spin quantum number (I) greater than zero.
- The transfer of energy occurs at a wavelength that coincides with the radio frequency.
- Also, energy is emitted at the same frequency when the spin comes back to its base level.
- Therefore, by measuring the signal which matches this transfer, the processing of the NMR spectrum for the concerned nucleus is yield.

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NMR Spectroscopy Applications

1. NMR spectroscopy is a Spectroscopy technique used by chemists and biochemists to investigate the properties of organic molecules, although it is applicable to any kind of sample that contains nuclei possessing spin. For example, the NMR can quantitatively analyze mixtures containing known compounds. NMR can either be used to match against spectral libraries or to infer the basic structure directly for unknown compounds.
2. Once the basic structure is known, NMR can be used to determine molecular conformation in solutions as well as in studying physical properties at the molecular level such as conformational exchange, phase changes, solubility, and diffusion.
3. Detection of aromaticity.
4. Detection of hydrogen bonding: Intermolecular hydrogen bonding shifts the absorption for a concerned proton downfield. The extent of hydrogen bonding varies with the solvent, concentration of the solution and the temperature. Intramolecular hydrogen bonding also shifts the absorption downfield. The two types of hydrogen bonding can be distinguished as the intramolecular hydrogen bonding is not concentration dependant.
5. Detection of electronegative atom or group. It is known that the presence of an electronegative atom or group in the neighbourhood of the proton cause deshielding and the signal is shifted downfield. Greater the electronegativity of the adjacent atom, smaller is the tau value of absorption for the concerned proton. Fluorine causes more downward shift as compared to oxygen and oxygen in turn causes more downward shift as compared to nitrogen and so on.
6. Importance in quantitative analysis. NMR spectroscopy is gaining importance for the Quantitative analysis of the compounds.

Difference between UV and IR Spectroscopy

The major difference between the UV and IR spectroscopy is that the UV spectroscopy is used to obtain the absorption spectra of a substance whereas the IR spectroscopy used to obtain functional groups present in a molecule.

Both ultraviolet and infrared spectroscopies are different forms of electromagnetic radiation, self-propagation waves that alternately formed with magnetic and electric fields. Although this basic similarity, there are several significant differences too.

The major differences are different from their wavelengths and frequencies. UV rays have a higher frequency and shorter wavelength than infrared. The wavelength of the IR is about 1 mm and 750 nm, although the ultraviolet wavelength is about 400 nm and 10 nm.

UV Spectroscopy: It has a shorter wavelength as compared to the visible light, higher energy, and frequency. It could break several organic bonds and cause skin cancer. (Changes in the level of electronic energy within the molecule).
IR Spectroscopy: It has a longer wavelength as compared to the visible light, thus low frequency and less energy per photon (Rotational and vibration movements of the molecule changes).

Difference between NMR and IR spectroscopy

Spectroscopy is the study of the interaction between electromagnetic radiation and matter. Depending on the energy of light employed, the sample contains different types of excitation and it serves as a means of classifying different spectroscopy methods.
IR gives information about the functional groups in a sample component in the range of 400-4000 cm−1. NMR spectroscopy used to study the absorption of radiofrequency radiation by nuclei in a magnetic field. This method uses radio waves.
FT-IR and NMR spectroscopy can give chemical information about sample molecules. The main difference between NMR and an IR spectroscopy is that the NMR spectroscopy is used to identify organic structure while FT-IR spectroscopy used to determine the presence and absence of functional groups in the sample.

Applications of Spectroscopy in various disciplines/fields

1. Spectroscopy is used as a tool for studying the structures of atoms and molecules.
2. Spectroscopy also provides a precise analytical method for finding the constituents in material having unknown chemical composition. In a typical spectroscopic analysis, a concentration of a few parts per million of a trace element in a material can be detected through its emission spectrum.
3. **Medical or Nursing Applications:**
   - UV lamps used to disinfect surgical operating rooms
   - Using MRI spectroscopy to detect tumors
   - A phone app that uses light reflection to help determine if a toddler has an eye tumor
4. **For Future CIA Agents/Spies:**
   - U.S. Spy Agencies Seek Tech to Identify Deadly Chemicals From 30 Meters Away
5. **Forensics:**
   - Anyone interested in Art Forgery
   - Using bloodstains at crime scenes to determine age of a suspect or victim
6. **Engineering:**
   - Quantum Engineering
   - Spectroscopy can be useful for engineers and material scientists
   - Civil engineers use light/ lasers to study icy roads
7. **Astronomy:**
   - In astronomy the study of the spectral emission lines of distant galaxies led to the discovery that the universe is expanding rapidly and isotropically (independent of direction).
8. **Agriculture/Dairy:**
   - Spectroscopy in agriculture/ farming

- Dairy/ Milk
- Grain fed beef or grass fed beef
- Fake or old beef products

9. **Computer Science/Engineering:**
   - The Basics of a Quantum Computer
     Water
10. **Career in the Outdoors:**
    - UV lamp used to disinfect water
    - Using Infrared Light to help manage wildlife
    - Environmental Analysis/ Science
    - Water Contamination
11. **Geology:**
    - Spectral Geology
    - Studying geological samples using high resolution spectroscopy(link is external)
12. **Ice cores:**
    - Laser based systems for gases and water isotopes from ice cores
13. **Physical and Occupational Therapy:**
    - Using MRI to help patients recover from stokes
    - Using IR spectroscopy to prevent ACL tears

Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC Accredited with 'A++' Grade.
Ph:07104-242919, 242623,242588 Website: www.ycce.edu, Email: principal@ycce.edu

Department of Applied Chemistry (2024-25)

Numericals

1. **A solution of trpyptophan has an absorbance at 280nm of 0.54 in a 0.5 cm cell. Given the absorption coefficient of tryptophan is 6.4 x 10^3 L cm^{-1} mol^{-1}. What is concentration of solution.**

Solution:

Given: x= 0.5 cm, A =0.54, ε = 6.4 x 10^3 L cm^{-1} mol^{-1}

A = εCx

$$C = \frac{A}{\varepsilon x} = \frac{0.54}{6.4 \times 10^3 \times 0.5} = 0.000168 \text{ M}$$

2. **Calculate the molar absorptivity of a 1 x 10^{-4} M solution which has an absorbance of 0.20, when the path length is 2.5 cm.**

Solution:

x=2.5cm, C=1 x 10^{-4} M, A =0.20

As A = εCx

$$\therefore \text{ molar absorptivity } (\varepsilon) = \frac{A}{Cx} = \frac{0.20}{1 \times 10^{-4} \times 2.5} = 800 \text{ dm}^3 \text{ cm}^{-1} \text{ mol}^{-1}$$

3. **Calculate the molar absorptivity of a 0.5 x 10^{-3} M solution which has an absorbance of 0.17 and the path length is 1.3 cm.**

Solution:

x=1.3cm, C=0.5 x 10^{-3} M, A =0.17

As A = εCx

$$\therefore \text{ molar absorptivity } (\varepsilon) = \frac{A}{Cx} = \frac{0.17}{0.5 \times 10^{-3} \times 1.3} = 261.53 \text{ dm}^3 \text{ cm}^{-1} \text{ mol}^{-1}$$

4. **A 1 x 10^{-4} M solution of an analyte when taken in a cell of 1 cm thickness shows absorbance is 0.139. Claculate the analytes molar absorptivity.**

Solution:

x=1cm, C=1 x 10^{-4} M, A =0.139

As A = εCx

$$\therefore \text{ molar absorptivity } (\varepsilon) = \frac{A}{Cx} = \frac{0.139}{1 \times 10^{-4} \times 1} = 1390 \text{ dm}^3 \text{ cm}^{-1} \text{ mol}^{-1}$$

5. **A $CaCO_3$ solution shows a transmittance of 90% when taken in a cell of 1.9 cm thickness. Calculate its concentration if molar absorption coefficient is 9000 dm^3 cm^{-1} mol^{-1}**

Solution:

x=1.9 cm, ε=9000 dm^3 cm^{-1} mol^{-1}

Absorbance $(A) = log\left(\frac{I_0}{I}\right)$ $log\left(\frac{100}{90}\right) = 0.045$

A = εCx

$$C = \frac{A}{\varepsilon x} = \frac{0.045}{9000 \times 1.9} = 2.631 \times 10^{-6} \text{ mol / dm}^3$$

6. **A solution of thickness 2 cm can transmit 40% of incident light. Calculate its concentration if molar absorption coefficient is 6000 $dm^3\ cm^{-1}\ mol^{-1}$**

Solution:

x=2 cm, ε=6000 $dm^3\ cm^{-1}\ mol^{-1}$

Absorbance $(A) = log\left(\frac{I_0}{I}\right)\ \ log\left(\frac{100}{40}\right) = 0.0398$

A = εCx

$$C = \frac{A}{\varepsilon x} = \frac{0.0398}{6000 \times 2} = 3.316 \times 10^{-5} \text{ mol / dm}^3$$

7. **A solution has a transmittance of 20% when taken in a cll of 2.5 cm thickness. Calculate its concentration if molar absorption coefficient is 12000 $dm^3\ cm^{-1}\ mol^{-1}$**

Solution:

x=2.5 cm, ε=12000 $dm^3\ cm^{-1}\ mol^{-1}$

Absorbance $(A) = log\left(\frac{I_0}{I}\right)\ \ log\left(\frac{100}{20}\right) = 0.698$

A = εCx

$$C = \frac{A}{\varepsilon x} = \frac{0.698}{12000 \times 2.5} = 2.33 \times 10^{-5} \text{ mol / dm}^3$$

8. **A solution containing 5g/L of a compound A in a 2 cm cell transmits 45 % of light at 410 nm. Calculate the molar absorption coefficient. Given molecular wt of A=70**

Solution:

Absorbance $(A) = log\left(\frac{I_0}{I}\right)\ \ log\left(\frac{100}{45}\right) = 0.3468$

Given C= 5g/L, MW = 70

C_M= C/MW = 5/70 = 0.07

As A = εCx

$$\therefore \text{ molar absorptivity } (\varepsilon) = \frac{A}{Cx} = \frac{0.03468}{0.07 \times 2} = 2.477 \text{ cm}^{-1}\text{ mol}^{-1}\text{ dm}^3$$

9. **A monochromatic light is incident on a solution of 0.05 molar concentration of an absorbing substance. The intensity of the radiation is reduced to one fourth of the initial value after passing through 10 cm length of the solution, calculate molar absorption coefficient.**

Solution:

According to Beers Lamberts law

Absorbance $(A) = log\left(\frac{I_0}{I}\right)$

But A = εCx

$\therefore$ log (I_o/I) = εCx

$$\therefore \log \frac{I_0}{I_0/4} = \varepsilon\ 0.05 \times 10$$

$$\varepsilon = 1.204\ \text{dm}^3\ \text{cm}^{-1}\ \text{mol}^{-1}$$